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DATA320 Writing Assignment 1

1. The presenter is elaborating on the field of data science from a point of view that breaks down its components for those who may be interested in it. Terms like machine learning, advanced analytics, and predictive analytics are given a stage for exploration that he uses to show how data can be used to extract knowledge and insights from a situation (Noisy data). These insights are then used as actionable means for a business or organization that stakeholders can invest in. The intersection of math/statistics, computer science, and business expertise comes together to make up the very essence of data science. The value that one gets out of data science methods goes hand in hand with the complexity of the problem the data is trying to address. Descriptive, diagnostic, predictive, and prescriptive analytics all exist within this conundrum. Describing and ascertaining why data presents itself the way it does before utilizing it for predictions and industry changes is imperative for reliable outcomes in the field. The baseline business and domain expertise that come prior to this process are also pivotal for success and subsequent data mining or processing. The exploration stage within this data science lifecycle leverages analytical tools that formulate the high-value tests and visuals that will empower the machine learning modeling. The level of computing power and high-quality data required for the model's actionable changes in the future is massive. It's also in one's best interest to

further visualize the outcomes and insights derived from the analysis and modeling as you go further in the cycle (revise and revisit as necessary). On an organizational scale, the roles that make up these tasks include business/data analysts, data engineers, and data scientists. Analysts come up with questions and visualizations for the insights to help answer the questions that the data is meant to answer. The engineers help find, clean, and explore the data before the data scientists explore, model, and visualize alongside the engineers and analysts. The slight overlap between the responsibilities of each role makes it critical to have further collaboration across each role. Business analysts sometimes might need to indulge in machine learning, or data scientists might end up needing to find the data on their own. Regardless, the core functions behind the field of data science remain the same.

2. This video is simply meant to inform viewers about the field of data science on a broad scale. From there, they can deeply investigate the specific concepts that may stand out to them. In the current age of technological advancement, it is extremely important for the next cycle of data experts to be well-versed in their work. Data is everywhere that a consumer of technology may look or find themselves indulging in. From the videos they view to the photos they take, data is stored and extracted by different individuals on a daily basis. Without an extensive grasp on what the data science lifecycle entails, the future generation of data scientists, engineers, and analysts could never hope to successfully cooperate and develop solutions to the problems that data works with or attempts to solve. It permeates almost every other medium that can be digitally explored (Healthcare, music, financial, editorial, etc.). IBM realizes this, and is doing us a justice by telling specific populations of interest what they're really getting into when they start

researching data science. It makes the journey more enticing for those truly earnest in the pursuit of this knowledge.

3. Although I knew a great deal about the field of data science and the components that go into the creation of a project that exists within the data science lifecycle, this video was extremely informative when it began talking about things from a business expertise perspective. It never really crossed my mind how much an organization would require its data scientists to consistently collaborate with data analysts and engineers prior to shipping out a solution that works for the stakeholders who are waiting on them. Familiarity with each step in the process is crucial, even if the initial job that is required of each of these roles does not state this. It means that you're slowly preparing yourself for unforeseen circumstances that the procedures will undoubtedly have. Each company is different, and by that same token, the responsibilities that each one may bestow on these roles will be different as well. However, what remains the same is the ability of the employee to adapt and utilize their domain expertise in situations that would place them at an advantage when developing proposals from insights.
4. The streamlined nature of the presentation style significantly improved the fluidity of my learning. It didn't feel like a chore to take notes on the necessary concepts from the video, as he broke them down in writing while speaking about them broadly. Doing these things in tandem meant that I could acquire the knowledge from both sources at an understandable pace. I find this to be especially important for a field as diverse and complex as data science, since even the basics can be overwhelming for a large majority of people who are new to computing, heavy math/statistics, and business. My presentation of similar concepts would certainly adopt the same presentation style prior to

delving into new concepts that explore the things he brought up in more detail. This is because I feel as though this is the best way for an unfamiliar audience to learn. Thrusting oneself into the brunt of data science without effectively breaking down the largest chunks into digestible information would be foolish. Like many STEM-related fields, it is much better to go piece by piece before bringing everything together (Paints a clearer picture).

5. Although this video did a great job at presenting high-level concepts and processes within the data science lifecycle in a simple enough manner for interested populations to understand, I would've liked to hear more about the concepts that go into machine learning and the analysis from it, as I'm already quite familiar with everything else. Pairing the practical nature of data mining, processing, and exploration alongside the slightly ambiguous and unforeseen nature of data modeling and algorithmic thinking is difficult for me sometimes. I've worked with different machine learning methods before, and even evaluated the effectiveness of supervised and unsupervised models to leverage their potential in related situations. However, I can't help but feel like I'm still not reaching the full extent of what those operations should really be. If IBM could've explained them from an organizational perspective with a similar approach used in this video, then I certainly would've enjoyed it a lot more. Things like natural language processing, Girvan-Newman's algorithm, and algorithmic relations to graph theory would've been extremely interesting to explore further.